

# Experiment 'p\_2\_4\_b\_difference' Results

June 3, 2026

**Experiment outcome:** SUCCESS

**Bad responses:** 0

**Responses containing assume :** 0

**Responses containing expect :** 0

**Resolution attempts:** 1

**Hard fails (resolution):** 0

**Soft fails (resolution):** 0

**Verification attempts:** 1

## Problem Specification

**Problem name:** p\_2\_4\_b\_difference

**Natural language statement:** Write a method that takes two integers and then returns the difference.

**Method signature:** p\_2\_4\_b\_difference(x: int, y: int) returns (result: int)

## Ensures

- `result == x - y`

## GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

### Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the difference.

The signature should be:

`p_2_4_b_difference(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x - y`

Produce and show only the Dafny body of this method, including the curly braces that  
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor  
→ 'expect' anywhere in your code.

### GenAI → Program

**System fingerprint:** null

**ID:** chatcmpl-DmdX4Z5MruoGpGxhwByMyCxV2LGfI

**Model:** gpt-5.5-2026-04-23

**Created at:** 1780484078

**Finish reason:** stop

**Usage:** 260 tokens in, and 22 tokens out

```
{  
  result := x - y;  
}
```

## Final Program

```
method p_2_4_b_difference(x: int, y: int) returns (result: int)  
  ensures result == x - y  
{  
  result := x - y;  
}
```

## Total Token Usage

**Input tokens:** 260

**Output tokens:** 22

**Reasoning tokens:** 0

**Sum of ‘total tokens’:** 282

## Experiment Timings

**Overall Experiment** started at 1780484077868, ended at 1780484083041, lasting 5173ms (5.17 seconds)

**Iteration #1** started at 1780484077869, ended at 1780484083040, lasting 5171ms (5.17 seconds)